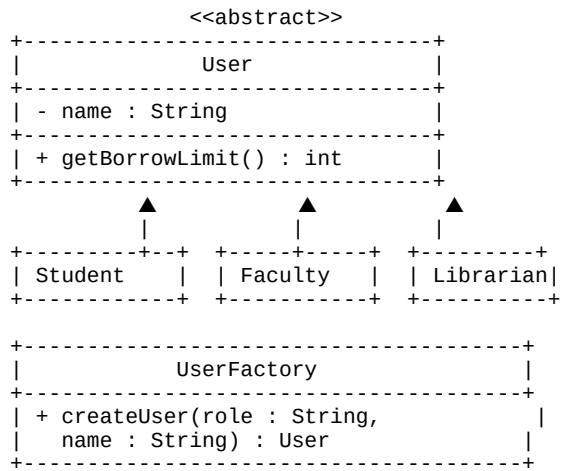


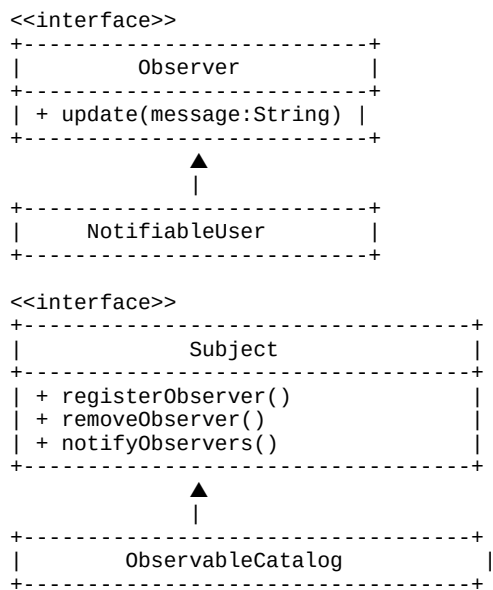
Singleton – LibraryCatalog

```
+-----+
|      LibraryCatalog      |
+-----+
| - instance : LibraryCatalog |
| - books   : List<Book>      |
+-----+
| - LibraryCatalog()         |
| + getInstance() : LibraryCatalog |
| + addBook(book : Book)      |
| + getBooks()  : List<Book>   |
+-----+
```

Factory – User Creation



Observer – Notification System



Builder – Book Construction

```
+-----+
|           Book           |
+-----+
| - title : String         |
| - authors : List<String> |
| - edition : String       |
| - genre : String         |
| - publisher : String     |
+-----+
```

```
+-----+
|           BookBuilder    |
+-----+
| + addAuthor() : BookBuilder |
| + edition() : BookBuilder   |
| + genre() : BookBuilder     |
| + publisher() : BookBuilder |
| + build() : Book            |
+-----+
```

BookBuilder -----> Book

Integrated – Complete System

LibraryCatalog -----> Book <----- BookBuilder

User



Student Faculty Librarian

UserFactory -----> User

ObservableCatalog -----> NotifiableUser